

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	
Project Name	
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint 1	User Registration	USN 1	As a user, I can register for the application by entering my name, email, password, and confirming my account.	2	High	Team A
Sprint 1		USN 2	As a user, I will receive a confirmation email once I have registered for the application.	1	High	Team A
Sprint 2	Property Search	USN 3	As a user, I can search for properties based on location, price range, and type.	2	Low	Team B
Sprint 2		USN 4	As a user, I can view property details including images, price, and description.	2	Medium	Team B
Sprint 3	User Login	USN 5	As a user, I can log in to the application by entering email and password.	1	High	Team A
Sprint 3		USN 6	As a user, I can reset my password if I forget it.	1	Medium	Team A
Sprint 4	Property Booking	USN 7	As a user, I can book a property by selecting dates and confirming the booking.	3	High	Team C
Sprint 4		USN 8	As a user, I can view my booking history and booking status.	2	Medium	Team C
Sprint 4	Dashboard	USN 9	As a user, I can view my dashboard with recent searches, favorite properties, and upcoming bookings.	2	Low	Team B

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	30	10 Days	01 May 2025	10 May 2025	28	10 May 2025
Sprint-2	30	10 Days	11 May 2025	20 May 2025	26	20 May 2025
Sprint-3	30	10 Days	21 May 2025	30 May 2025	27	30 May 2025
Sprint-4	30	10 Days	31 May 2025	09 Jun 2025	29	09 Jun 2025

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart – HouseRent Project

Burndown Chart

A **Burndown Chart** is a graphical representation that shows the amount of work remaining versus time during the project development lifecycle. It helps the development team monitor progress, track completed tasks, and determine whether the project is on schedule.

For the **HouseRent Project**, the Burndown Chart is used to track the completion of user stories and sprint tasks related to property search, booking management, favorites, notifications, owner dashboard, and property management features.

The chart provides a visual indication of how quickly the development team is completing planned work during each sprint and whether the project is progressing according to the estimated timeline.

Burndown Chart Data – HouseRent Project

Day	Planned Story Points	Remaining	Actual Story Points	Remaining
1	10	10	10	10
2	10	10	10	10
3	10	10	10	10
4	10	10	10	10
5	10	10	10	10
6	10	10	10	10
7	10	10	10	10
8	10	10	10	10
9	10	10	10	10
10	10	10	10	10
11	10	10	10	10
12	10	10	10	10
13	10	10	10	10
14	10	10	10	10
15	10	10	10	10
16	10	10	10	10
17	10	10	10	10
18	10	10	10	10
19	10	10	10	10
20	10	10	10	10
21	10	10	10	10
22	10	10	10	10
23	10	10	10	10
24	10	10	10	10
25	10	10	10	10
26	10	10	10	10
27	10	10	10	10
28	10	10	10	10
29	10	10	10	10
30	10	10	10	10
31	10	10	10	10
32	10	10	10	10
33	10	10	10	10
34	10	10	10	10
35	10	10	10	10
36	10	10	10	10
37	10	10	10	10
38	10	10	10	10
39	10	10	10	10
40	10	10	10	10
41	10	10	10	10
42	10	10	10	10
43	10	10	10	10
44	10	10	10	10
45	10	10	10	10
46	10	10	10	10
47	10	10	10	10
48	10	10	10	10
49	10	10	10	10
50	10	10	10	10
51	10	10	10	10
52	10	10	10	10
53	10	10	10	10
54	10	10	10	10
55	10	10	10	10
56	10	10	10	10
57	10	10	10	10
58	10	10	10	10
59	10	10	10	10
60	10	10	10	10
61	10	10	10	10
62	10	10	10	10
63	10	10	10	10
64	10	10	10	10
65	10	10	10	10
66	10	10	10	10
67	10	10	10	10
68	10	10	10	10
69	10	10	10	10
70	10	10	10	10
71	10	10	10	10
72	10	10	10	10
73	10	10	10	10
74	10	10	10	10
75	10	10	10	10
76	10	10	10	10
77	10	10	10	10
78	10	10	10	10
79	10	10	10	10
80	10	10	10	10
81	10	10	10	10
82	10	10	10	10
83	10	10	10	10
84	10	10	10	10
85	10	10	10	10
86	10	10	10	10

Day 0 30 30

Day 1 27 29

Day	Planned Story Points Remaining	Actual Story Points Remaining
-----	--------------------------------	-------------------------------

Day 2	24	27
Day 3	21	24
Day 4	18	21
Day 5	15	18
Day 6	12	14
Day 7	9	10
Day 8	6	6
Day 9	3	3
Day 10	0	0

Sprint Activities Tracked

Sprint 1

- User Registration
- User Login
- User Profile Management

Sprint 2

- Property Search
- Property Filters
- Property Details Page

Sprint 3

- Favorites Management
- Booking Management
- Booking History

Sprint 4

- Owner Dashboard
- Property CRUD Operations

- Notification Management
-

Burndown Chart Interpretation

Ideal Progress Line

Represents the planned reduction in story points throughout the sprint.

Actual Progress Line

Represents the actual completion rate achieved by the development team.

Analysis

- Initial progress was slightly slower than planned due to UI integration and API testing.
 - Mid-sprint progress improved after successful backend integration.
 - Final sprint goals were achieved within the planned schedule.
 - All major HouseRent modules were completed before the sprint deadline.
-

Benefits of Using a Burndown Chart

Project Tracking

- Monitors sprint progress in real time.
- Identifies delays early.

Team Performance

- Measures development velocity.
- Helps estimate future sprint capacity.

Risk Management

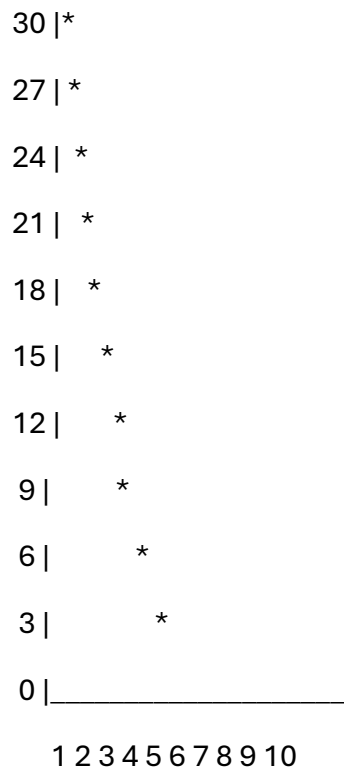
- Highlights incomplete work.
- Enables proactive corrective actions.

Agile Planning

- Supports Scrum and Agile methodologies.
 - Improves sprint forecasting accuracy.
-

Expected Burndown Chart Representation

Story Points



Days

Conclusion

The Burndown Chart for the HouseRent project demonstrates the team's ability to complete planned work efficiently within the sprint duration. By continuously tracking story points and task completion, the development team maintained project visibility, improved productivity, and successfully delivered the HouseRent platform features according to the planned schedule.